# MARTIAL WORLD

A Mini Project Report

submitted by

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to the APJ Abdul Kalam Technological University
in partial fulfilment of the requirements for the award of the Degree

of

Master of Computer Applications

**Department of Computer Applications**

MES College of Engineering

Kuttippuram, Malappuram – 679582

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# Declaration

I undersigned hereby declare that the project report “MARTIAL WORLD” submitted for partial fulfilment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under supervision of Mr.Balachandran KP, Associate Professor, Department of Computer Applications. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Saran S Krishnan

(MES24MCA2049)

06/10/2025

DEPARTMENT OF COMPUTER APPLICATIONS

MES COLLEGE OF ENGINEERING, KUTTIPPURAM

# CERTIFICATE

This is to certify that the report entitled **MARTIAL WORLD** is a bonafide record of the Mini Project work during the year 2025-26 carried out by S**ARAN S KRISHNAN(MES24MCA2049)** submitted to the APJ Abdul Kalam Technological University, in partial fulfilment of the requirements for the award of the Master of Computer Applications, under my guidance and supervision. This report in any form has not been submitted to any other University or Institution for any purpose.

Internal Supervisor

Head of The Department

# Acknowledgment

I express my sincere gratitude to our principal Dr.Rahumathuniza I for permitting and providing all facilities to carryout this project. I would like to express my sincere gratitude to my internal supervisor, Mr. Balachandhran KP, for his invaluable guidance, support, and encouragement throughout the development of this mini project. I also extend my thanks to the Head of the Department Prof. Hyderali K and all faculty members of the Department of Computer Applications, MES College of Engineering, for providing the necessary resources and a conducive environment for learning and innovation.

I am deeply grateful to my fellow martial artists and peers who inspired the vision behind *Martial World*. Their insights and experiences shaped the direction of this project.

Finally I thank my family and friends for their unwavering support and motivation. Above all, I extend my heartfelt gratitude ot Almighty for giving me the strength and wisdom to complete this project.

Saran S Krishnan

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# Abstract

Martial World is a pioneering web application designed to unify the global martial arts community through a dedicated digital ecosystem. It addresses the fragmentation in existing platforms by integrating commerce, social networking, and personal progress tracking tailored specifically for martial artists. The app offers a curated marketplace for gear and nutrition, a discipline-specific media-sharing network, and tools for tracking training, belt promotions, and competition history.

Built using Javascript for cross-platform development and Firebase for backend services, Martial World leverages Firebase and AWS for real-time updates and secure hosting. This report outlines the motivation, objectives, system architecture, and implementation methodology of the project, culminating in a robust solution that empowers martial artists to grow, connect, and thrive.

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# Chapter 1. Introduction

Martial arts is not just a sport—it is a philosophy, a discipline, and a lifestyle embraced by millions worldwide. Despite its rich cultural heritage and widespread practice, the digital infrastructure supporting martial artists remains fragmented. Existing platforms often dilute martial arts content with unrelated material, making it difficult for practitioners to find authentic gear, track progress, or engage with like-minded individuals.

*Martial World* aims to bridge this gap by offering a unified platform exclusively for martial artists. It combines e-commerce, social networking, and personal development tools into a single application. The platform is designed to support practitioners across disciplines like Karate, Taekwondo, Muay Thai, Jiu-Jitsu, and Kung Fu.

The project falls under the domain of full-stack web and mobile development, with a focus on user-centric design and scalable architecture. The UI/UX is crafted to be intuitive, visually engaging, and responsive across devices, ensuring accessibility for users of all skill levels.

The problem addressed is the lack of a dedicated digital space for martial artists. Existing solutions are either too generic or fail to integrate commerce, community, and progress tracking effectively. *Martial World* offers a holistic solution that celebrates martial arts culture and supports practitioners' growth.

## 1.1 Motivation

It has been found that the martial artists have personally experienced the challenges of navigating multiple platforms to meet training, gear, and community needs. The absence of a unified digital space that understands and caters to the unique lifestyle of martial artists inspired the creation of *Martial World*. This project is not just a technical endeavor—it is a personal mission to empower martial artists through technology.

The motivation stems from the desire to build a platform that reflects the values of discipline, growth, and community inherent in martial arts. By integrating commerce, content, and progress tracking, *Martial World* aims to simplify and enrich the journey of every practitioner.

## 1.2 Objectives

The primary objective of *Martial World* is to develop a mobile application that:

- Empowers users to track and analyze their martial arts journey.
- Fosters a vibrant, discipline-specific social network.
- Centralizes martial arts-related commerce through a curated marketplace.

## 1.3 Contributions

Key contributions of *Martial World* include:

- A dedicated e-commerce hub for martial arts gear and nutrition.
- A social media platform focused solely on martial arts content.
- A progress tracker for training, belt promotions, and competition history.
- Integration of Firebase for real-time updates and AWS for secure media storage.

These contributions address the limitations of existing systems and offer a comprehensive solution tailored to the martial arts community.

## 1.4 Report Organization

The report is organized in such a way that the chapter 2 gives the overview of the existing system and the proposed solution, chapter 3 describes the development approach tools used and the details of the modules that are designed, chapter 4 summerizes the outputs, various UI/UX designed and alterations on it, finally it summerizes the project archievements and the future scope in chapter 5.

# Chapter 2. System Study

Martial World is designed to address the fragmented digital experience faced by martial artists. The system aims to unify commerce, community, and personal progress tracking into a single, discipline-specific platform.

## 2.1 Existing System

Presently, martial artists rely on multiple platforms to fulfill their needs—e-commerce sites for gear, social media for community engagement, and spreadsheets or journals for tracking progress. These systems are generic and not tailored to martial arts, leading to inefficiencies, lack of personalization, and diluted content. For example, platforms like Instagram or Amazon do not offer martial arts-specific filters or features. This fragmentation hinders growth and community cohesion.

Improving upon these limitations, Martial World proposes a unified solution that integrates all essential features into one seamless experience.

## 2.2 Proposed System

The proposed system, *Martial World*, is a mobile-first application that centralizes martial arts commerce, social networking, and progress tracking. It offers:

- A curated marketplace for gear, nutrition, and event tickets.
- A media-sharing platform focused solely on martial arts content.
- A personal dashboard for tracking training, belt promotions, and competition history.

Technologically, the system uses React Native for cross-platform development, Django REST for backend services, and Firebase for real-time updates. It is designed to be scalable, secure, and user-friendly.

Target users include martial artists of all levels, instructors, academies, and vendors specializing in martial arts products.

## 2.3 Functionalities of Proposed System

- **E-Commerce Hub:**
  - Marketplace for gear, apparel, weapons, nutrition, and event tickets.
  - Vendor onboarding and discipline-specific product categorization.
  - Secure payment integration with Stripe and Razorpay.
- **Entertainment & Social Network:**
  - Media sharing: photos, videos, reels, and stories.
  - Follower system, hashtags, and content discovery.
  - Personal branding tools for fighters, instructors, and academies.
- **Progress Tracker:**
  - Training logs and belt promotion records.
  - Goal-setting and fitness analytics.
  - Journal entries and performance insights.

# Chapter 3. Methodology

## 3.1 Introduction

To develop *Martial World*, an Agile methodology was adopted. Agile promotes iterative development, continuous feedback, and adaptive planning—ideal for building user-centric applications like this one. The project was divided into sprints, each focusing on specific modules such as e-commerce, social networking, and progress tracking. This approach ensured flexibility, faster delivery, and better alignment with user needs.

## 3.2 Software Tools

| | |
|---|---|
| Operating System | Windows 11 |
| Front End | JavaScript, HTML, CSS etc.. |
| Back End | Firebase |
| Database | Firebase |
| IDE | Visual Studio Code |
| Version Control | Git |

Table 3.2: List of software tools or languages used for the project development

## 3.3 Module Description

The system is divided into modular components to ensure maintainability and scalability.

### 3.3.1 User Authentication Module

This model handles the signup and signin of the users. Ensures login for only registered users.

### 3.3.2 Progress Tracker Module

This module empowers users to log their martial arts journey. They can choose martial arts of their choice and can start their journey. Each can focus on their own martial arts and can train with the syllabus already provided in the page and also they can set timer for each section. They can also choose their level and start working according to their belt level.

## 3.4 User Story

| **Role** | **Goal** | **Benefit** |
|---|---|---|
| Martial Artist | Buy gear and track progress | Saves time and improves focus |
| Instructor | Share content and manage students | Builds reputation and engagement |
| Vendor | Sell martial arts products | Reaches targeted audience |

Table 3.4: User Story

## 3.5 Product Backlog

| **ID** | **Name** | **Priority** | Estimate(hrs) | **Status** |
|---|---|---|---|---|
| 1 | User Registration | High | 5 | Completed |
| 2 | Login | High | 3 | Completed |
| 3 | Progress Tracker | High | 20 | Completed |
| 4 | Media Sharing | Medium | 20 | In Progress |
| 5 | Product Listing | Low | 20 | Planned |

Table 3.5: Product Backlog

## 3.6 Sprint Backlog

| **Sprint** | **Task** | **Status** |
|---|---|---|
| 1 | Firebase Auth Setup | Completed |
| 2 | UI Design for Login/Register | Completed |
| 3 | Product API Integration | Completed |
| 4 | Media Upload Feature | In Progress |

Table 3.6: Sprint Backlog

## 3.7 Project Plan

| User Story ID | Task Name | Start Date | End Date | Duration (Days) | Status |
|---|---|---|---|---|---|
| 1 | Sprint 1 | 14/07/2025 | 16/07/2025 | 3 | Complete |
| 2 | | 17/07/2025 | 19/07/2025 | | Complete |
| 3 | | 20/07/2025 | 23/07/2025 | | Complete |
| 4 | Sprint 2 | 24/07/2025 | 30/07/2025 | 7 | Complete |
| 5 | | 31/07/2025 | 06/08/2025 | | Complete |
| 6 | Sprint 3 | 07/08/2025 | 13/08/2025 | 7 | Complete |
| 7 | | 14/08/2025 | 21/08/2025 | | Complete |
| 8 | Sprint 4 | 22/08/2025 | 28/08/2025 | 7 | Planned |
| 9 | | 29/08/2025 | 30/08/2025 | | Planned |

Table 3.7: Project Plan

## 3.8 Database Design

The database schema includes collections of User details like Username, Email address.

| Email | Provider | Created | Signed In | UID |
|---|---|---|---|---|
| sd@gmail.com | Email | Oct 5, 2023 | Oct 5, 2023 | g1f8s8a1u7m... |

| | | | | |
|---|---|---|---|---|
| sd@gmail.com | Email | Oct 5, 2023 | Oct 5, 2023 | 1g1f8s8a1u7... |
| sd@gmail.com | Email | Oct 5, 2023 | Oct 5, 2023 | 2g1f8s8a1u7... |
| sd@gmail.com | Email | Oct 5, 2023 | Oct 5, 2023 | 3g1f8s8a1u7... |
| t1@gmail.com | Email | Oct 5, 2023 | Oct 5, 2023 | 4g1f8s8a1u7... |

Table 3.8: Database Design

This table summarizes user authentication data for the MariaWorld-MVP Firebase project. It lists registered users along with their email addresses, authentication provider (Email), account creation and last sign-in dates, and unique Firebase-generated UIDs. All users were created and signed in on October 5, 2023, suggesting a coordinated onboarding event. The UID ensures each user is uniquely identifiable, even when email addresses repeat. This table supports backend integration, user management, and secure access control within the application.

| **Field Name** | **Value** |
|---|---|
| Document ID | 1Hs5CNhEYPE... |
| createdAt | September 23, 2023 at 5:41:58 PM UTC-3 |
| email | t@gmail.com |
| username | tm |

Table 3.8: Database Design

# Chapter 4. Results and Discussions

## 4.1 Results

The *Martial World* application was successfully developed and tested across its core modules. Below are key screenshots and descriptions of the implemented features:

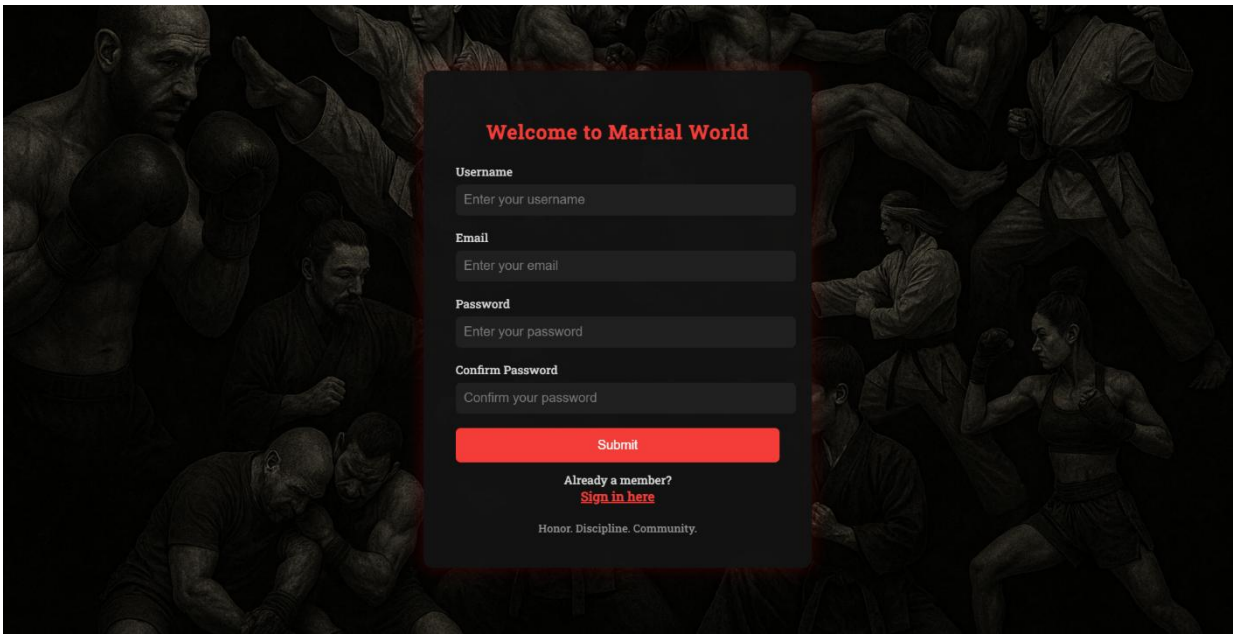



**Figure 4.1:** SignUp Page

This is the welcome screen of the app, featuring a clean UI with options to log in, register. It sets the tone for a martial arts-focused experience.

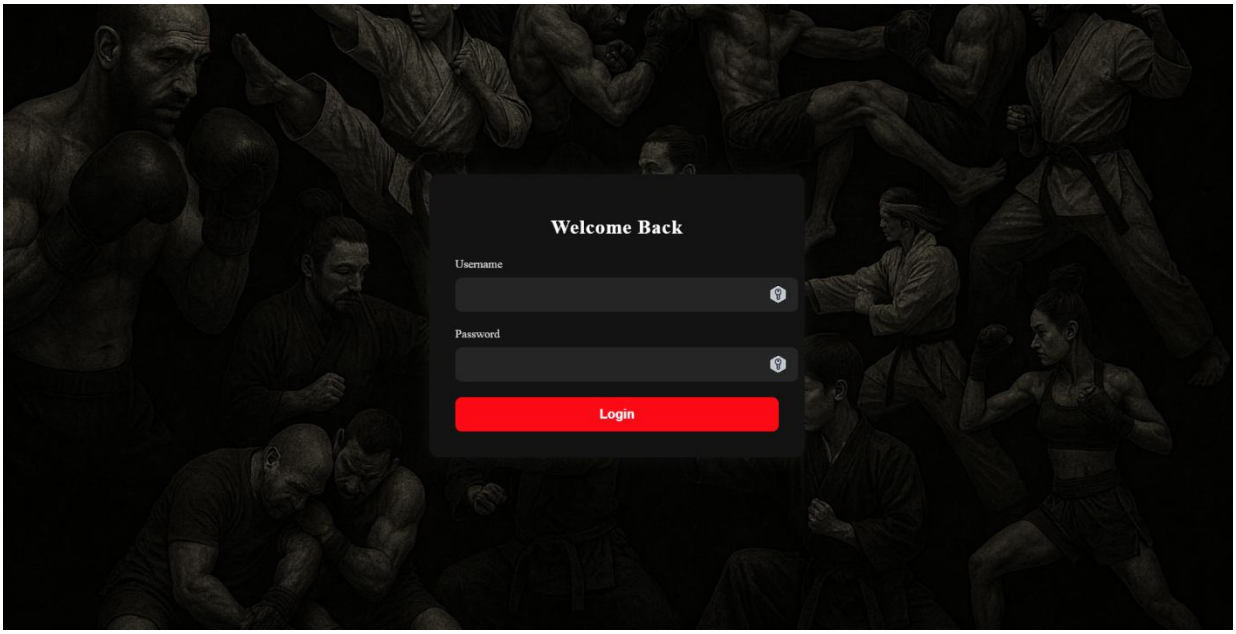



**Figure 4.2:** SignIn Page

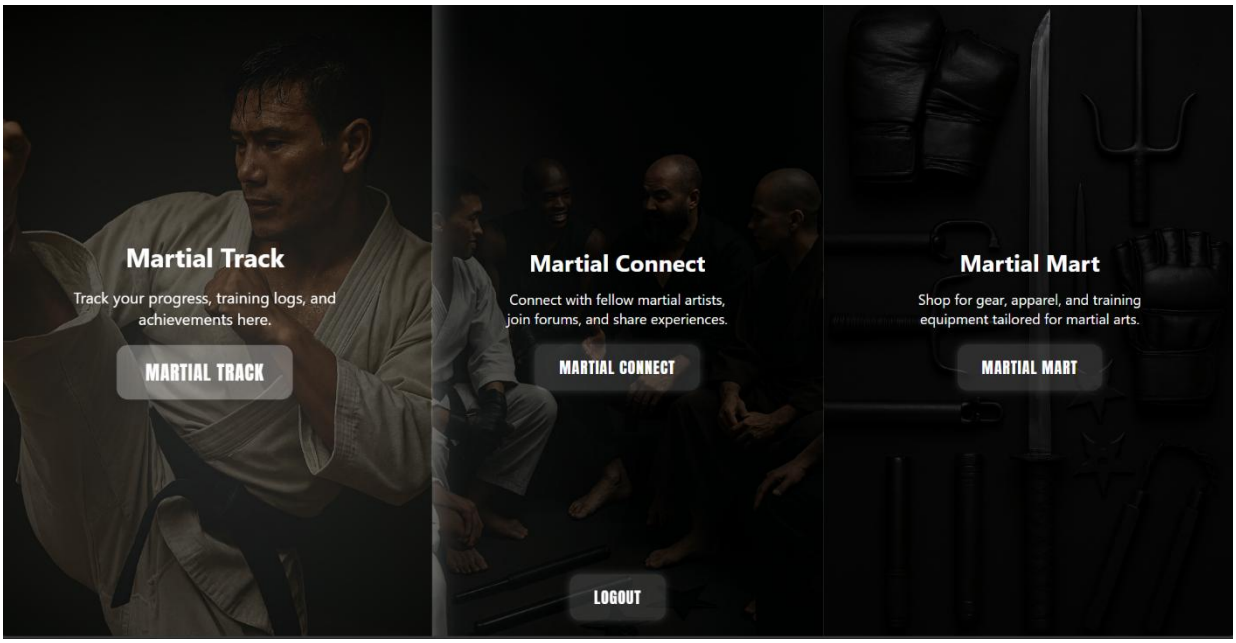



**Figure 4.3:** Dashboard Page

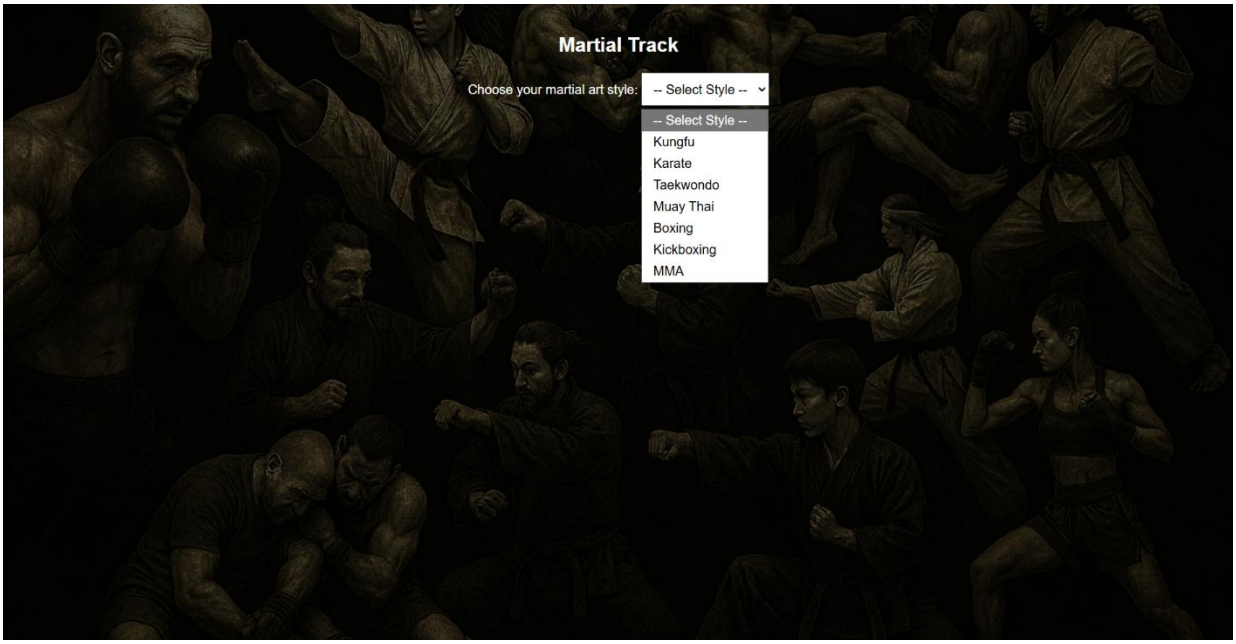



**Figure 4.4:** Martial Track Page

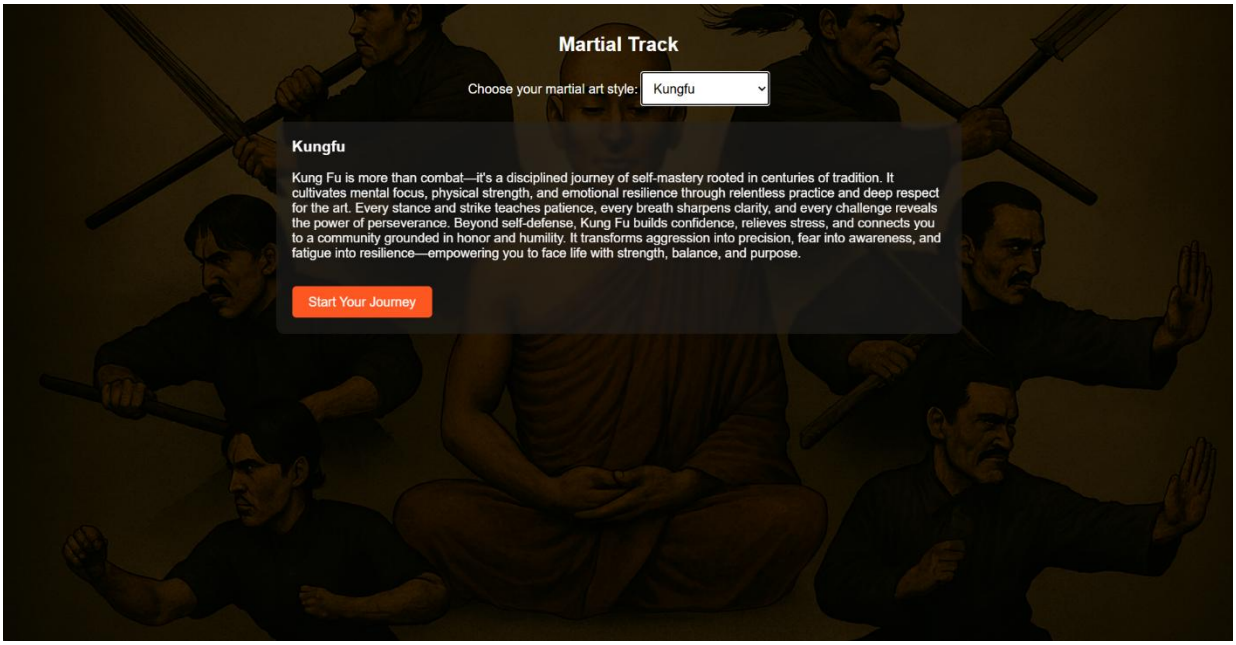



**Figure 4.5:** Kungfu Page

**Figure 4.6:** Kungfu Tracking Page (Belt Levels)

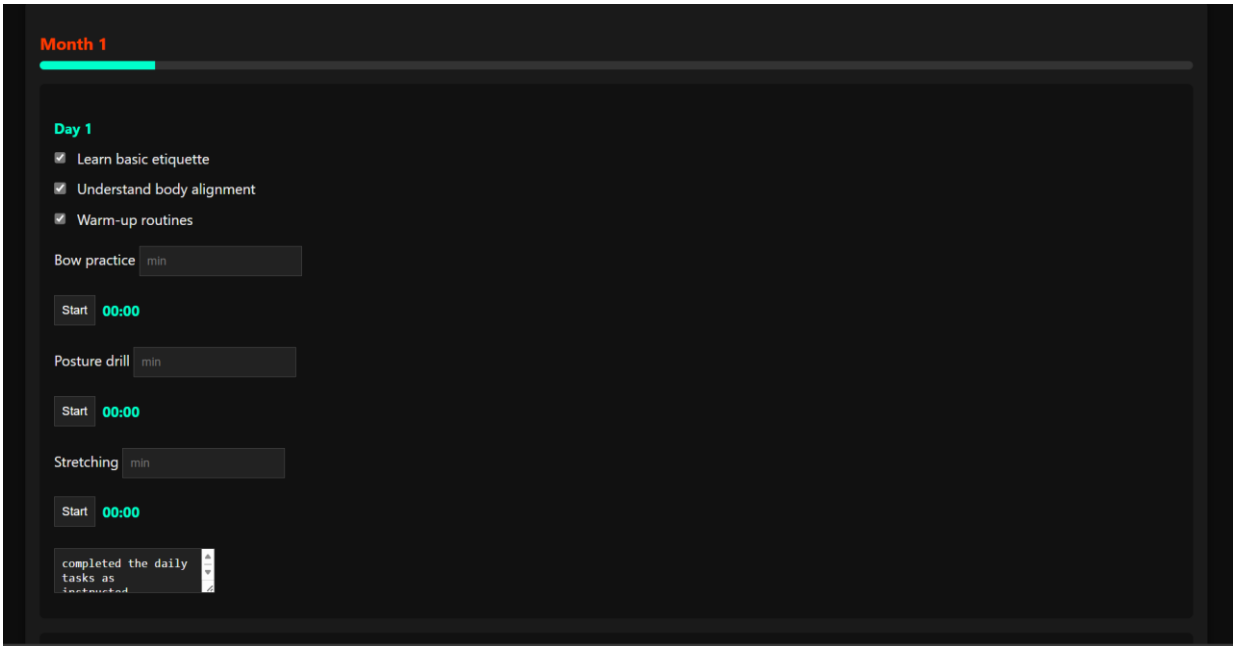



**Figure 4.6.1:** Kungfu Tracking Page (Timer and Checkbox)

# Chapter 5. Conclusion and Future Scope

*Martial World* is a comprehensive web application designed to unify the martial arts community through a dedicated digital ecosystem. It successfully integrates commerce, social networking, and personal development tools tailored to martial artists.

Key achievements include:

- A discipline-specific marketplace for gear and nutrition.
- A vibrant social platform for sharing martial arts content.
- A robust progress tracker for training, belt promotions.

The project demonstrates the power of full-stack development in solving real-world problems. While the MVP covers core functionalities, future enhancements could include creating own syllabus and academy management tools.

Limitations faced included time constraints for advanced features like live streaming and vendor analytics. These can be addressed in future iterations with extended development cycles and user feedback.

# References

[1] Firebase, “Firebase Documentation,” Google LLC, [Online]. Available: https://firebase.google.com/docs. . [Accessed: 01 Oct 2025].

[2] WAKO India Kickboxing Federation, “National Kickboxing Guidelines,” [Online]. Available: https://wakoindia.org/. . [Accessed: 01 Oct 2025].

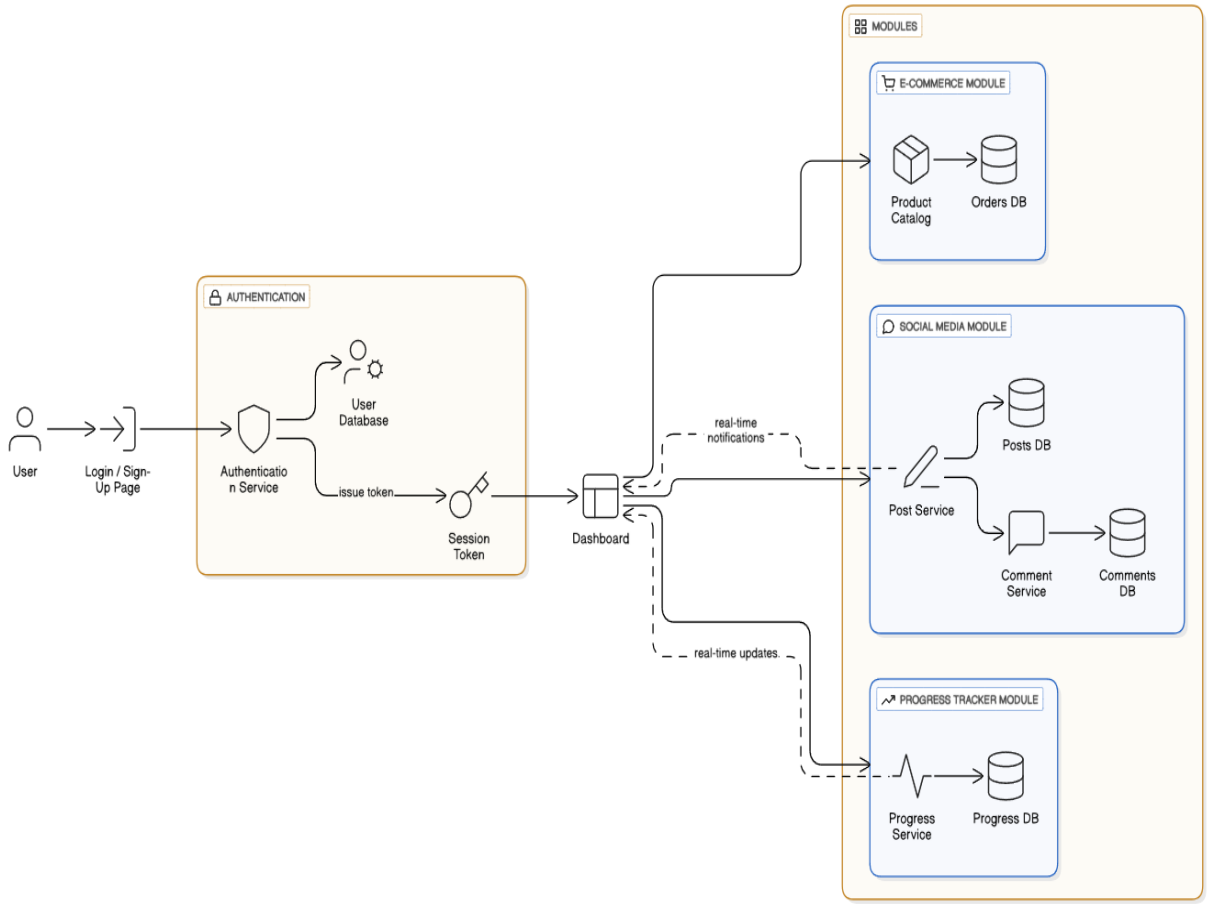
[3] "Getting started with Firebase on the Web" A concise walkthrough of setting up Firebase, including authentication, Firestore, and hosting. Ideal for web-based apps.

[4] HTML, CSS, and Javascript in 30 minutes A fast-paced crash course that introduces all three technologies in a single session—great for quick revision.

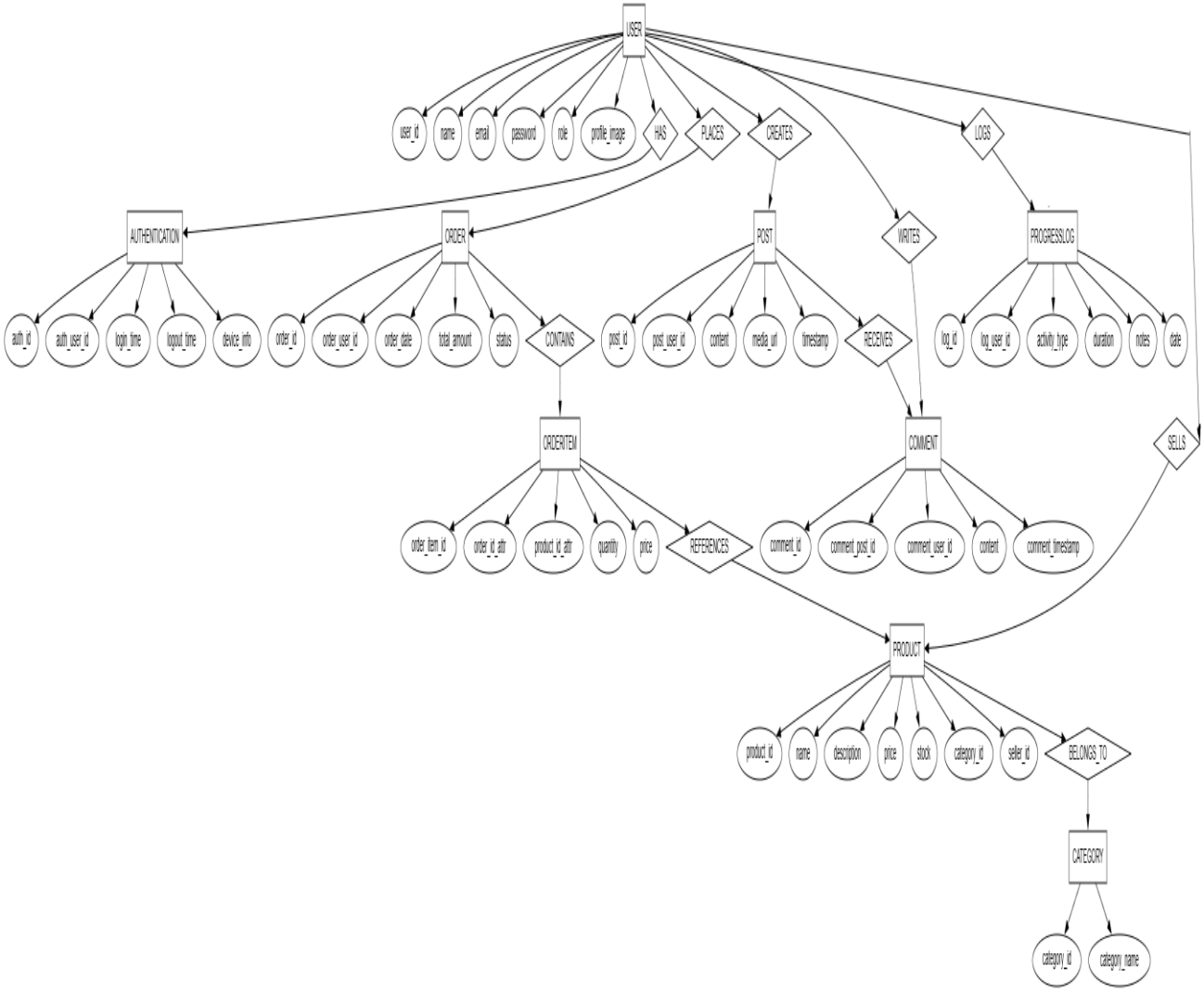
[5] Microsoft Copilot, AI Companion for Technical and Creative Assistance. [Online]. Available: https://copilot.microsoft.com. [Accessed: 06 Oct 2025].

# Appendix

## Appendix A Data Flow Diagram



# Appendix B ER Diagram



# Appendix C Source Code

## 1. Sign Up Page.

```
<body>
  <div class="login-container">
    <h2>Welcome to Martial World</h2>
    <form id="loginForm">
      <label for="username">Username</label>
      <input type="text" id="username" placeholder="Enter your username" required />
      <label for="email">Email</label>
      <input type="email" id="email" placeholder="Enter your email" required />
      <label for="password">Password</label>
      <input type="password" id="password" placeholder="Enter your password" required />
      <label for="confirmPassword">Confirm Password</label>
      <input type="password" id="confirmPassword" placeholder="Confirm your password"
required />
      <button type="submit">Submit</button>
      <div style="text-align: center; margin-top: 10px;">
        <span style="color: #ccc; font-size: 0.9rem;">Already a member?</span>
        <br />
        <a href="signin.html" style="color: #e53935; text-decoration: underline; font-weight:
bold;">Sign in here</a>
      </div>
    </form>
    <div class="footer">Honor. Discipline. Community.</div>
  </div>
</body>
```

## 2. Sign In Page

```
<body>
  <div class="login-container">
    <h2>Welcome Back</h2>
    <form id="signinForm">
      <label for="username">Username</label>
      <input type="text" id="username" name="username" required />
      <label for="password">Password</label>
      <input type="password" id="password" name="password" required />
      <button type="submit">Login</button>
    </form>
  </div>
 </body>
```

## 3. Dashboard

```
<body>
  <div class="section track">
    <div class="section-content">
      <h2>Martial Track</h2>
      <p>Track your progress, training logs, and achievements here.</p>
      <button onclick="location.href='martial-track.html'">Martial Track</button>
    </div>
  </div>
  <div class="section connect">
    <div class="section-content">
```

```
        <h2>Martial Connect</h2>
        <p>Connect with fellow martial artists, join forums, and share experiences.</p>
        <button onclick="location.href='martial-connect.html'">Martial Connect</button>
      </div>
    </div>
    <div class="section mart">
      <div class="section-content">
        <h2>Martial Mart</h2>
        <p>Shop for gear, apparel, and training equipment tailored for martial arts.</p>
        <button onclick="location.href='martial-mart.html'">Martial Mart</button>
      </div>
    </div>
    <div id="mouse-light"></div>
    <div class="logout-container">
    <button onclick="logout()">Logout</button>
    </div>
</body>
```

## 4. Martial Track Menu

```
  <body>
    <div class="overlay">
      <h2>Martial Track</h2>
      <form id="styleForm">
        <label for="martialArt">Choose your martial art style:</label>
        <select id="martialArt" required>
          <option value="">-- Select Style --</option>
          <option value="Kungfu">Kungfu</option>
```

```
        <option value="Karate">Karate</option>
        <option value="Taekwondo">Taekwondo</option>
        <option value="Muay Thai">Muay Thai</option>
        <option value="Boxing">Boxing</option>
        <option value="Kickboxing">Kickboxing</option>
        <option value="MMA">MMA</option>
      </select>
    </form>
    <div id="descriptionBox">
      <h3 id="styleTitle"></h3>
      <p id="styleDescription"></p>
      <button id="nextButton">Start Your Journey</buton>
    </div>
  </div>
</body>
```

## 5. Progress Tracker Bar

```
const progressPercent=totalGoals >0 ? Math.round((completedGoals / totalGoals) * 100) : 0;
const progressFill = monthDiv.querySelector(`#progress-${belt}-${monthIndex}`);
if (progressFill) {
  progressFill.style.width = `${progressPercent}%`;
}
```

## 6. Timer

```
function startTimer(button) {
  const exerciseDiv = button.closest(".exercise");
  const input = exerciseDiv.querySelector(".timeInput");
```

```
  const display = exerciseDiv.querySelector(".timer-display");
  let minutes = parseInt(input.value);
  if (isNaN(minutes) || minutes <= 0) {
    alert("Enter a valid number of minutes.");
    return;
  }
  let seconds = minutes * 60;
  button.disabled = true;
  display.textContent = formatTime(seconds);
  const interval = setInterval(() => {
    seconds--;
    display.textContent = formatTime(seconds);
    if (seconds <= 0) {
      clearInterval(interval);
      button.disabled = false;
      display.textContent = "Time's up!";
      playAlarm();
      exerciseDiv.style.border = "2px solid #ff3c00";
    }
  }, 1000);
}
function formatTime(sec) {
  const m = Math.floor(sec / 60);
  const s = sec % 60;
  return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`
}
```

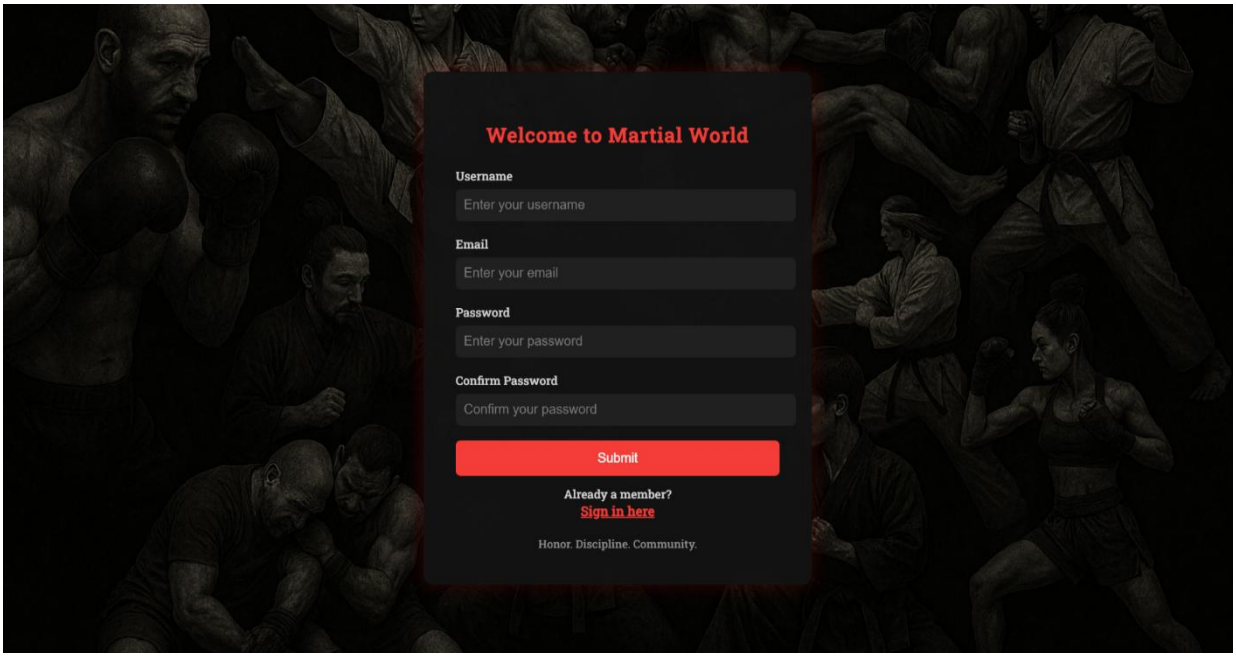
```
function playAlarm() {
  const audio = new Audio("https://actions.google.com/sounds/v1/alarms/alarm_clock.ogg");
  audio.play();
}
updateSyllabus(); // Load default on page load
```

## 7. Choosing Level

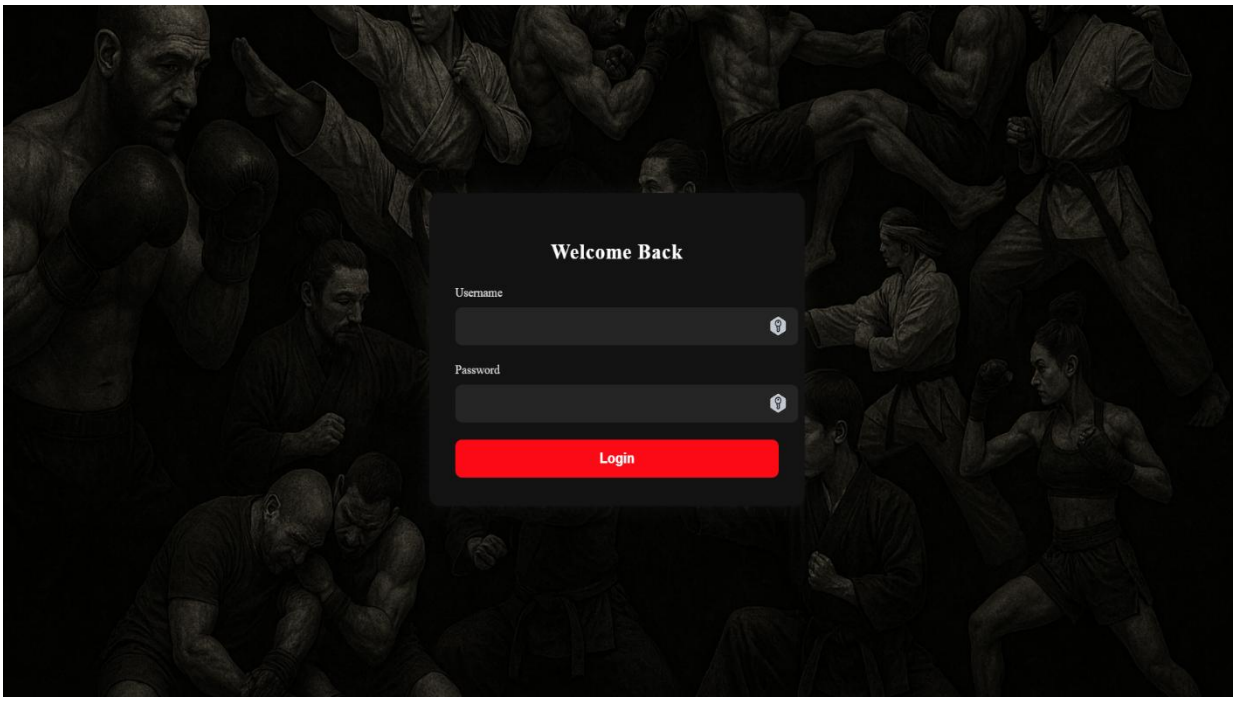
```
function updateSyllabus() {
  const belt = document.getElementById("belt").value;
  const content = document.getElementById("beltContent");
  const guide = document.getElementById("syllabusText");
  const data = syllabusData[belt];
  guide.innerHTML = `
    <strong>Focus for ${belt.charAt(0).toUpperCase() + belt.slice(1)} Belt:</strong><br/>
    <ul>
      <li><span style="color:#00ffcc;">Warm-up (5–10 min):</span> Stretching, breathing, light cardio</li>
      <li><span style="color:#00ffcc;">Technique (15–20 min):</span> Practice key moves like <em>${data.exercises.join(', ')}</em></li>
      <li><span style="color:#00ffcc;">Conditioning (10–15 min):</span> Strength drills, endurance work</li>
      <li><span style="color:#00ffcc;">Reflection (5 min):</span> Write notes, review goals, track progress</li>
    </ul>
    <em>Tip:</em> Stay consistent. Mastery comes from repetition, discipline, and honest effort.
`;
```

# Appendix D Screenshot

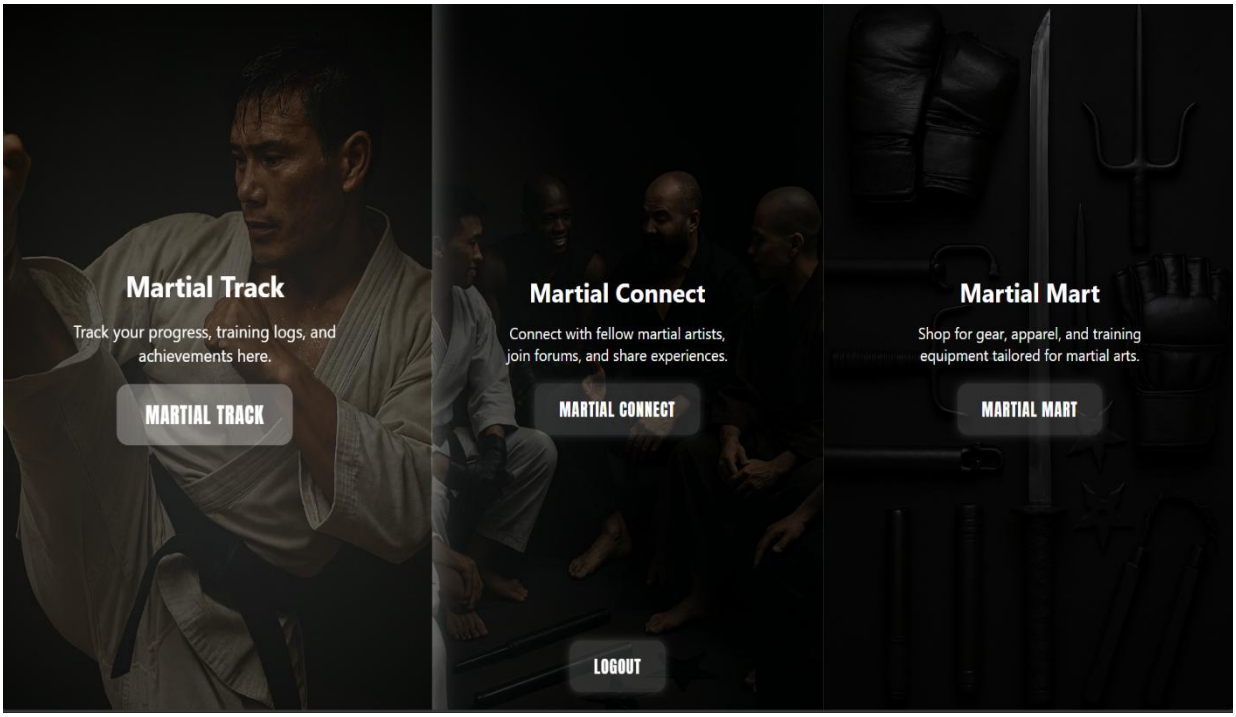
## 1. Sign Up Page



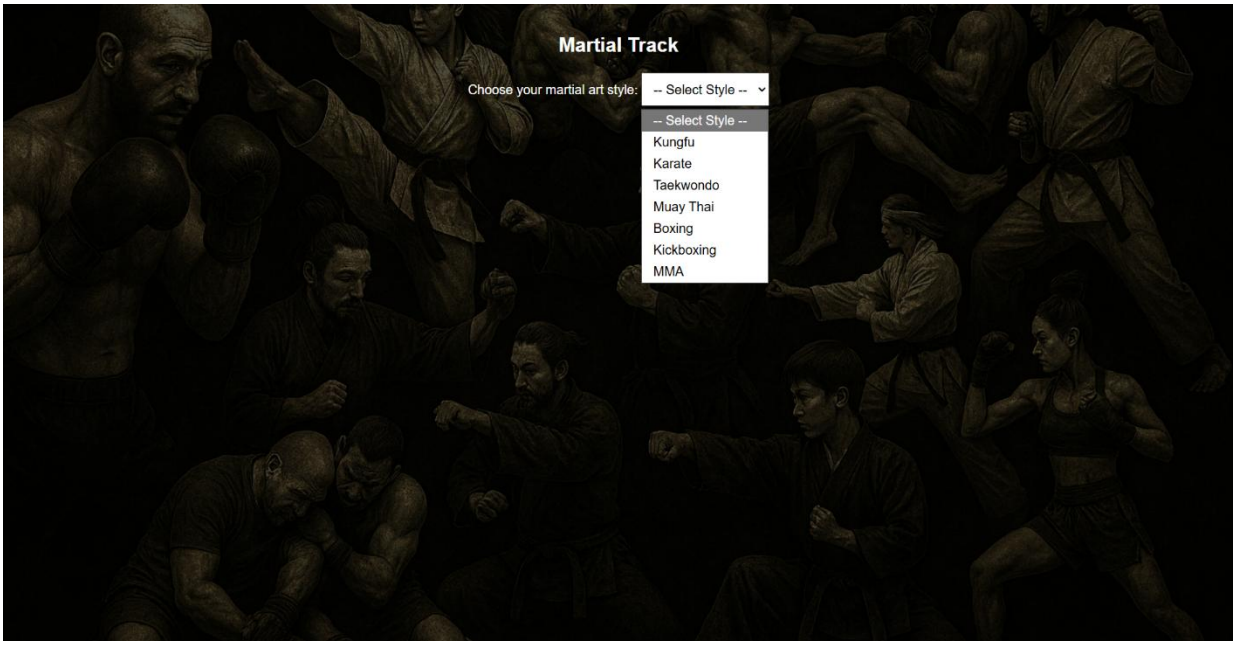

## 2. Sign In Page



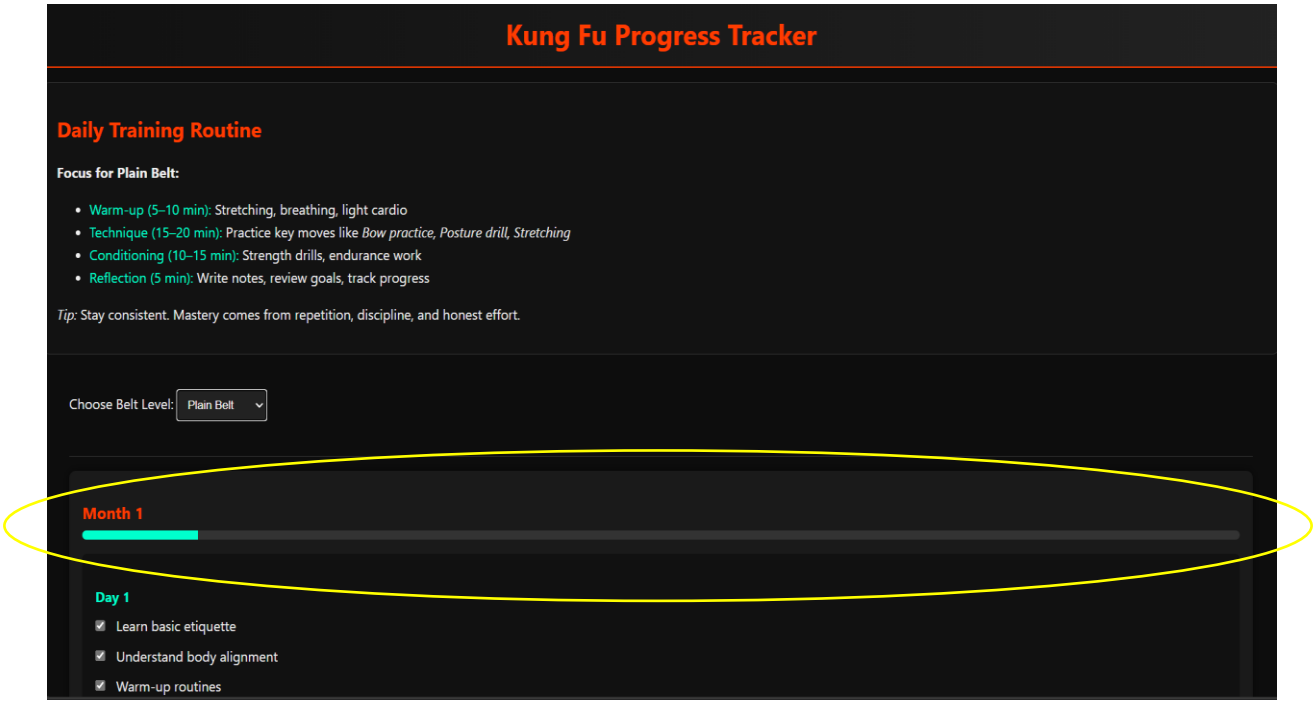
## 3. Dashboard



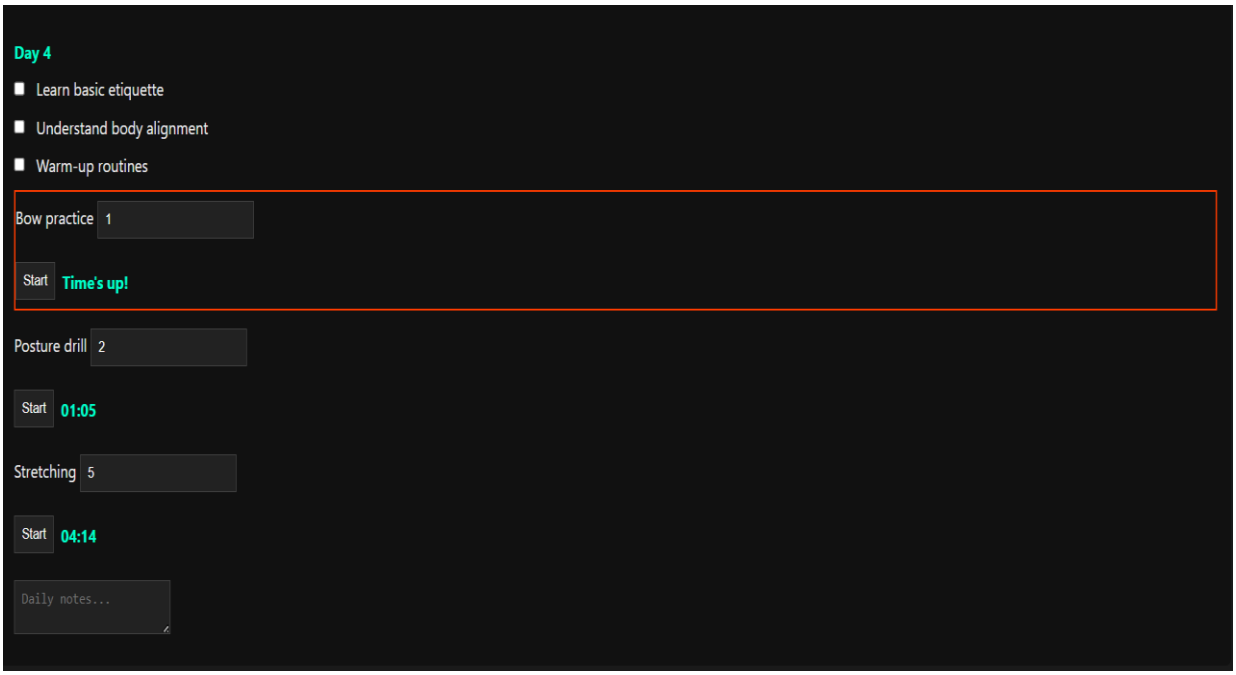

## 4. Martial Track Menu



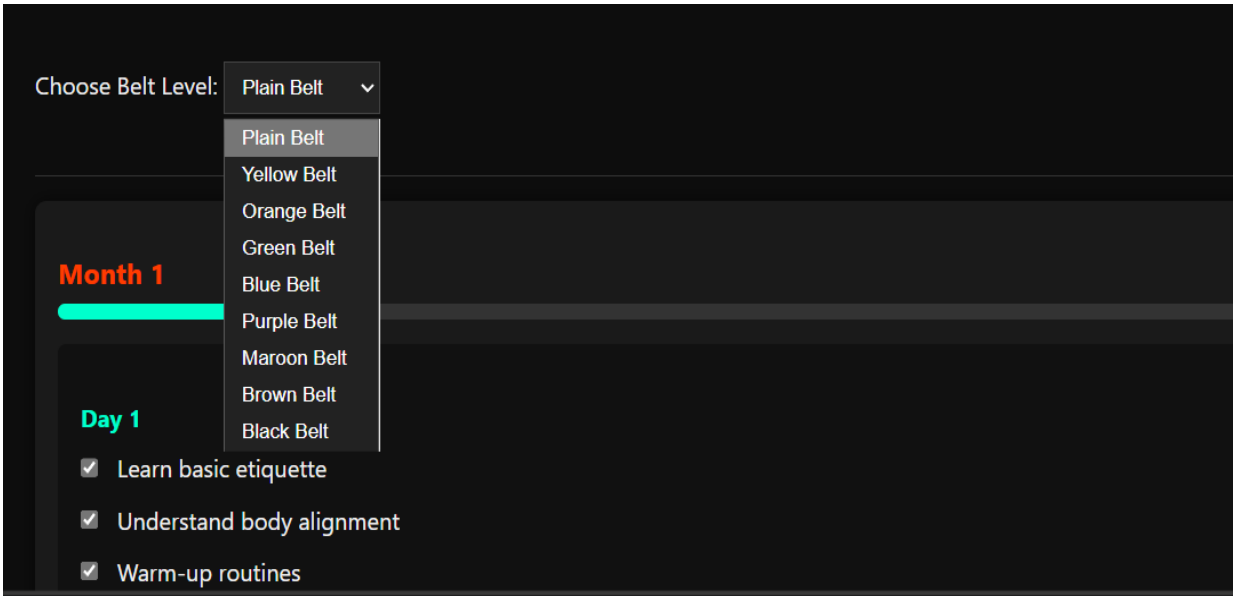
## 5. Progress Tracker Bar



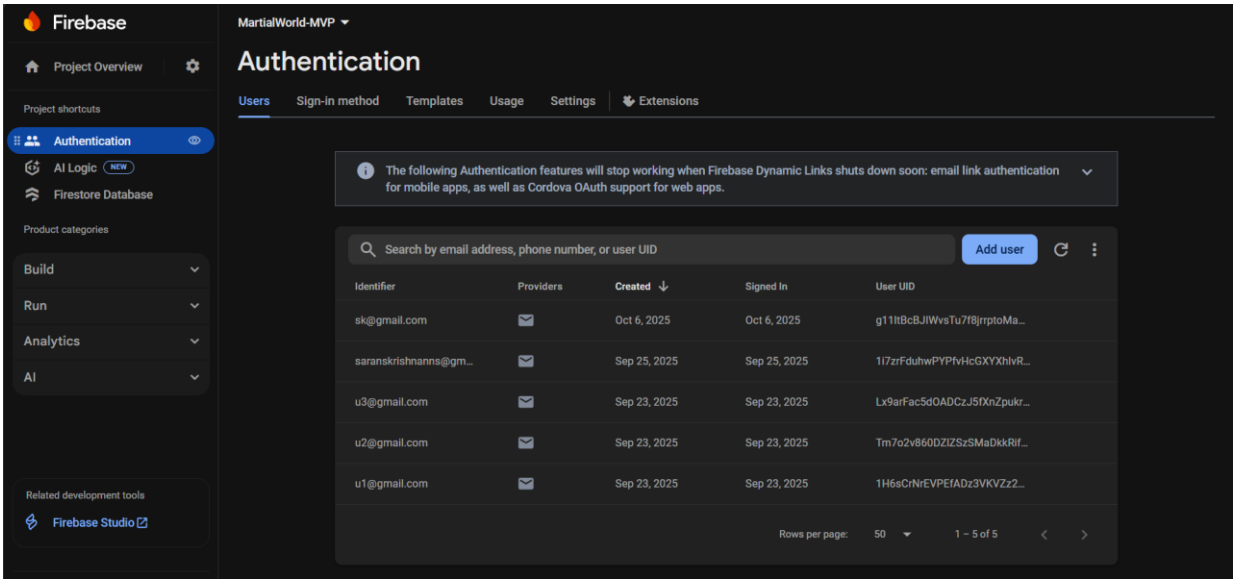

## 6. Timer



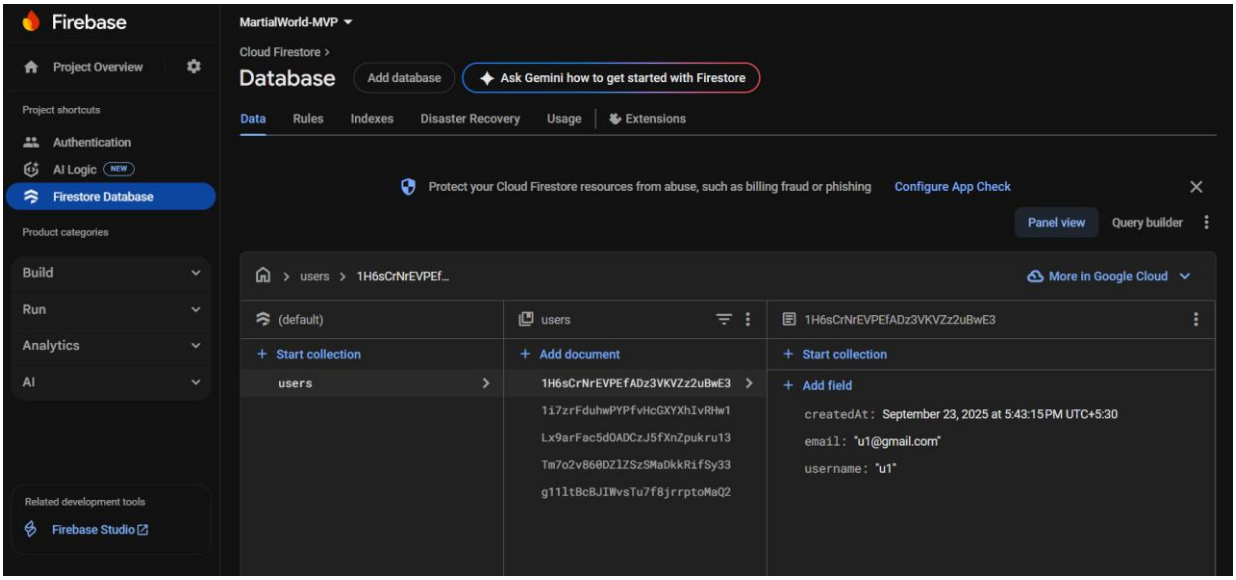
## 7. Choosing Level




## 8. Authentication



# 9. Database


Firebase
MartialWorld-MVP
Project Overview
Project shortcuts
Authentication
AI Logic NEW
Firestore Database
Product categories
Build
Run
Analytics
AI
Related development tools
Firebase Studio
Cloud Firestore >
Database
Add database
Ask Gemini how to get started with Firestore
Data
Rules
Indexes
Disaster Recovery
Usage
Extensions
Protect your Cloud Firestore resources from abuse, such as billing fraud or phishing
Configure App Check
Panel view
Query builder
users > 1H6sCrNrEVPEf...
More in Google Cloud
(default)
Start collection
users
users
Add document
1H6sCrNrEVPEfADz3VKVZz2uBwE3
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Start collection
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